Analytical Geometry & Calculus1 (MTH111)

Nile University

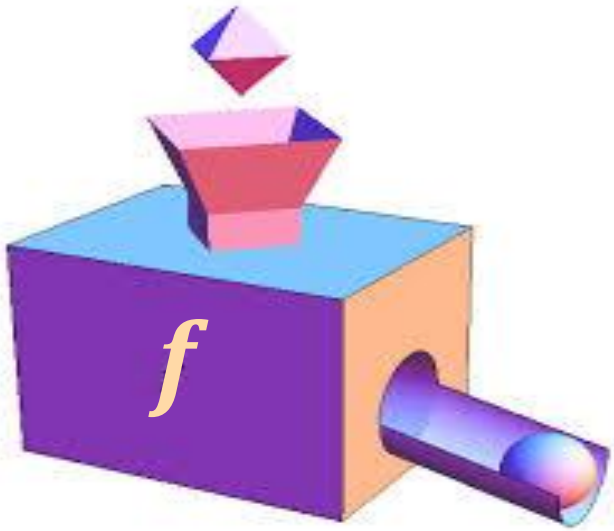
Lecture #2

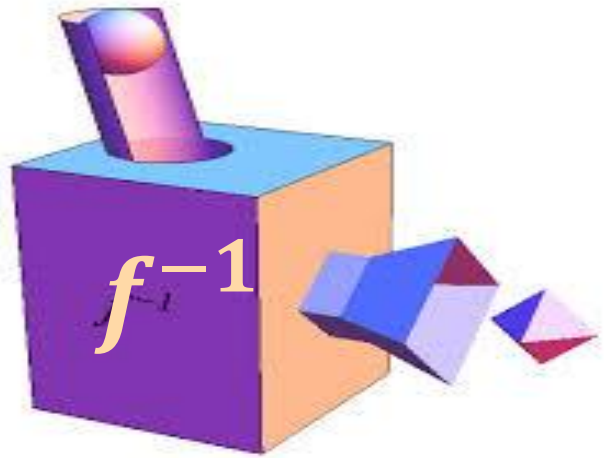
Revision on Functions

- **Inverse Function**
- **Exponential and Logarithmic Functions**
- **Derivatives**

Inverse Function

Introduction





Inverse: backward direction

Inverse Function

Introduction

let $y = 2x - 3$ **y in terms of x**

Rewrite the function such that x in terms of y

$$y = 2x - 3 \quad \Rightarrow 2x = y + 3 \quad \Rightarrow x = \frac{y+3}{2}$$

Function Notation:

$$f(x) = 2x - 3$$

Inverse Notation:

$$f^{-1}(x) = \frac{x+3}{2}$$

Inverse Function

Example: **Find the Inverse Function for** $f(x) = 2x^3 - 1$

$$y = 2x^3 - 1 \qquad y + 1 = 2x^3 \qquad \frac{y+1}{2} = x^3$$

$$x = \sqrt[3]{\frac{y+1}{2}} \qquad f^{-1}(x) = \sqrt[3]{\frac{x+1}{2}}$$

Steps to Find Inverse(till now)

[1] Replace $f(x)$ with y

[2] Find x in terms of y

[3] Replace x by $f^{-1}(x)$ and replace y by x

Inverse Function

Example: **Find the Inverse Function for**

$$f(x) = \frac{2x}{x-1}$$

$$y = \frac{2x}{x-1}$$

$$y(x-1) = 2x$$

$$yx - y = 2x$$

$$yx - 2x = y$$

$$x(y-2) = y$$

$$x = \frac{y}{(y-2)}$$

$$f^{-1}(x) = \frac{x}{(x-2)}$$

Inverse Function

Example: Find the Inverse Function for

$$f(x) = x^2 - 1, \quad x \geq 0$$

$$y = x^2 - 1$$

$$x^2 = y + 1$$

$$x = \pm\sqrt{y+1}$$

Ignore the negative

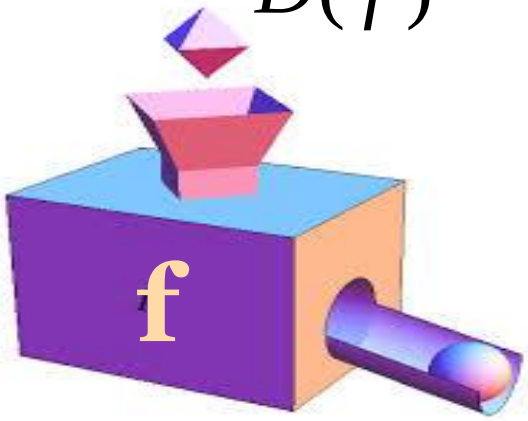
$$x = +\sqrt{y+1}$$

$$f^{-1}(x) = \sqrt{x+1}$$

Inverse Function

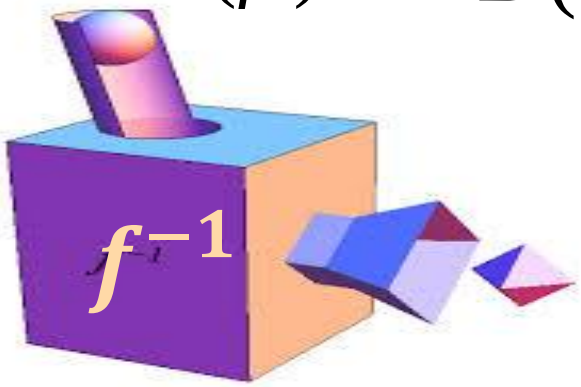
Domain and range of inverse function

$D(f)$



$R(f)$

$R(f) = D(f^{-1})$



$D(f) = R(f^{-1})$

- **Important Notes:-**
  - $D(f) = R(f^{-1})$:
  - $D(f^{-1}) = R(f)$:

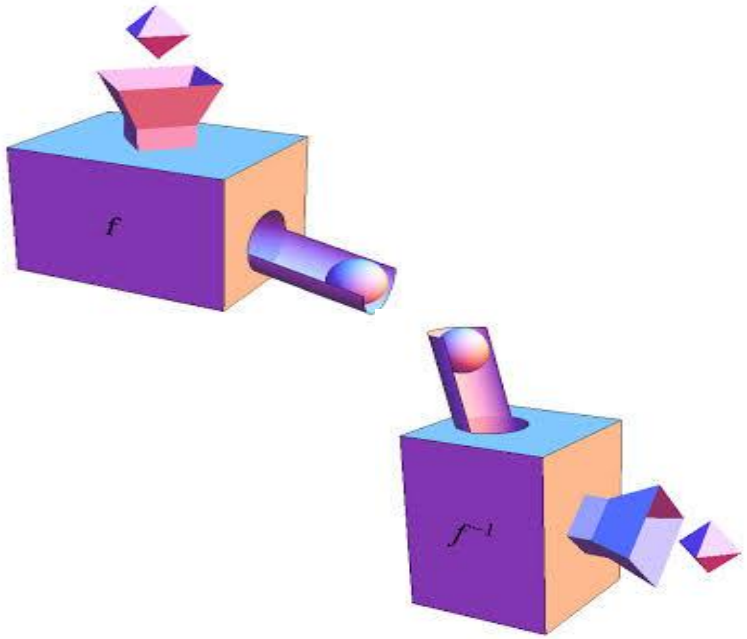
Domain and range are switched

Inverse Function

Cancellation equations

$$f\,(f^{-1}(x)) = x$$

$$f^{-1}(f\,(x)) = x$$



Inverse Function

Example: **Verify the cancellation equations**

$$f(x)=5x+10 \qquad f^{-1}(x)=\frac{1}{5}x-2$$

a)Check $f(f^{-1}(x))$

$$f(f^{-1}(x))=5\left(\frac{1}{5}x-2\right)+10$$

$$=x-10+10$$

$$=x$$

b)Check $f^{-1}(f(x))$

$$f^{-1}(f(x))=\frac{1}{5}(5x+10)-2$$

$$f^{-1}(f(x))=x+2-2$$

$$=x$$

One-to-One Functions

Example

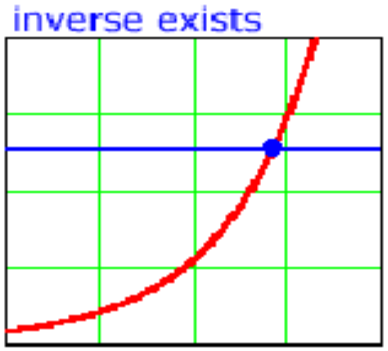
$y = x + 1$	$y = x^2$	$y^2 = x$
$x = 1 \rightarrow y = 2$	$x = 1 \rightarrow y = 1$	$x = 4 \rightarrow y = 2$ or -2
$x = 3 \rightarrow y = 4$	$x = -1 \rightarrow y = 1$	
One-to-One function	Not One-to-One function	Not a Function

One-to-One function A function that has distinct values at distinct elements in its domain is called one-to-one.

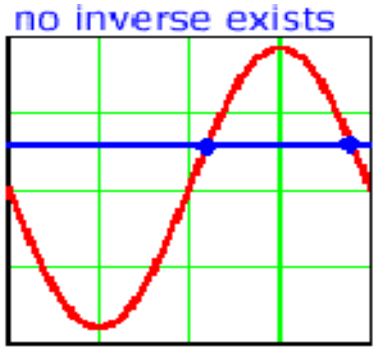
Only a one-to-one function can have an inverse

One-to-one Function

Check for One-to-One functions using The Horizontal Line Test



A One-to-One function

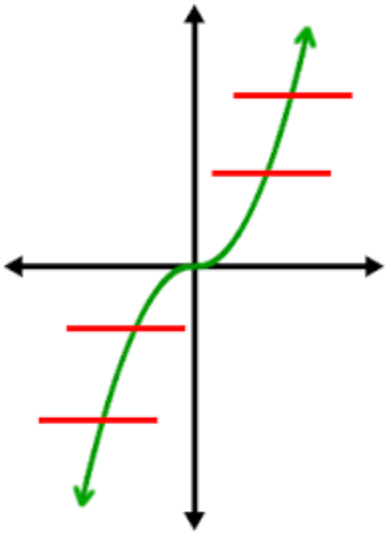


Not a One-to-One function

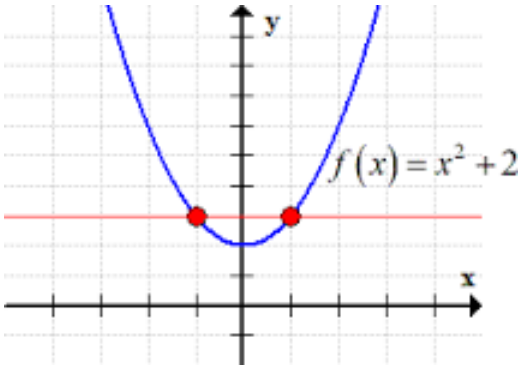
If the horizontal line intersects the curve in only one point, then the function is One-to-One

One-to-one Function

Check for One-to-One functions using The Horizontal Line Test

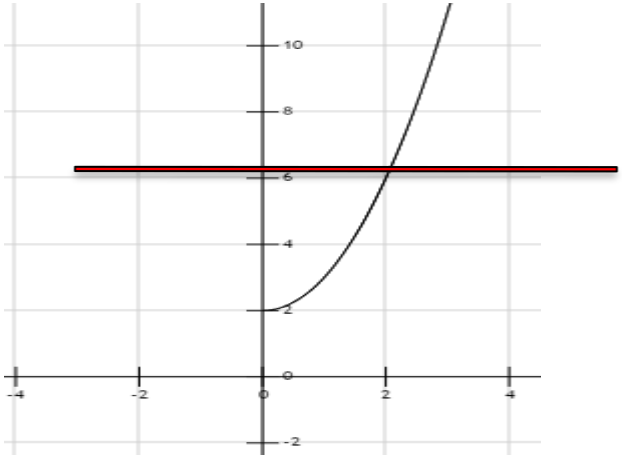


A One-to-One function



Not a One-to-One function

$$f(x)=x^2+2, x \succ 0$$



A One-to-One function

Finding Inverses

Steps to find the inverse function

[1] Check if $f(x)$ is one-to-one

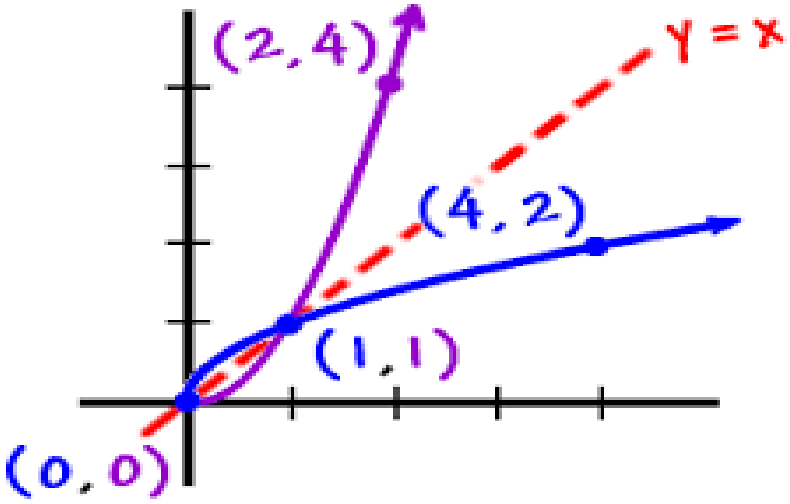
[2] Replace $f(x)$ with y

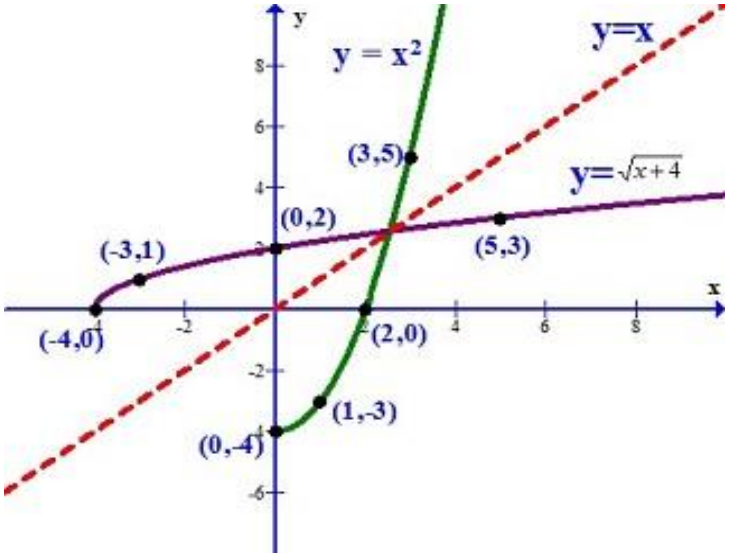
[3] Find x in terms of y

[4] Replace x by $f^{-1}(x)$ and replace y by x

Sketching the Inverse Function

Reflect about the line $y = x$

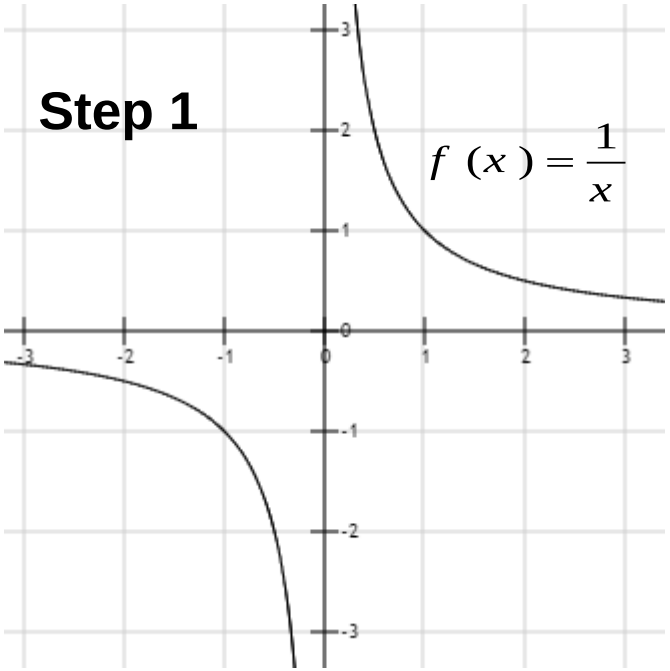




Finding Inverses: Example

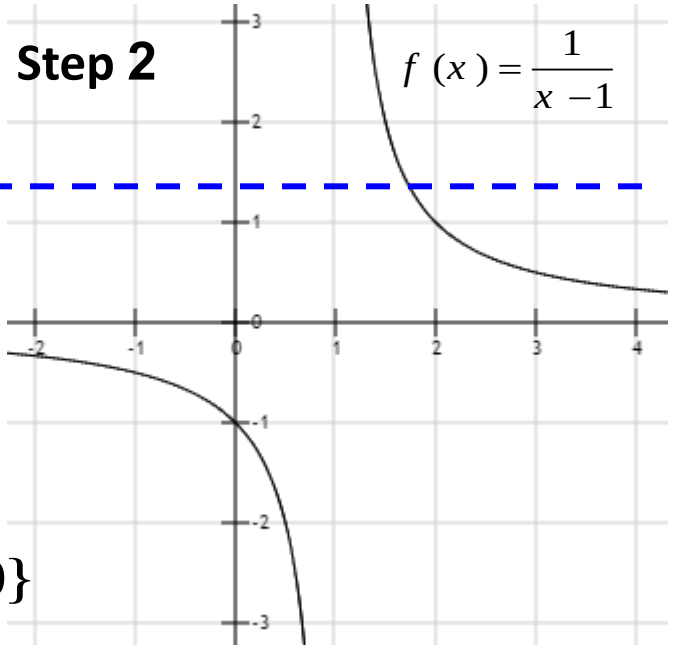
Example Find the inverse function (if it exists) of $f(x) = \frac{1}{x-1}$, sketch it , and write its domain and range

Step 1



The inverse function exists

Step 2



$$D(f^{-1}) = \Re - \{0\}$$

$$R(f^{-1}) = \Re - \{1\}$$

Finding Inverses: Example Continued

Example **Find the inverse function (if it exists) of $f(x)=\frac{1}{x-1}$, sketch it , and write its domain and range**

$$y = \frac{1}{x-1}$$

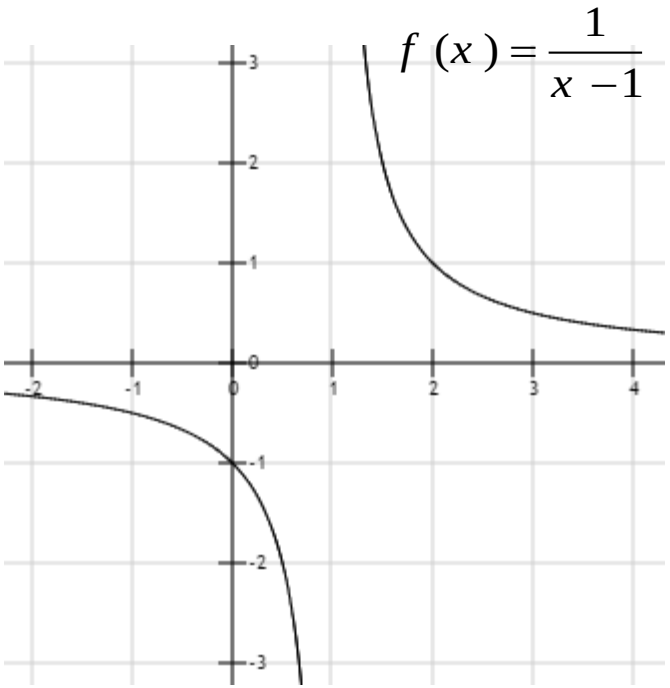
$$x - 1 = \frac{1}{y}$$

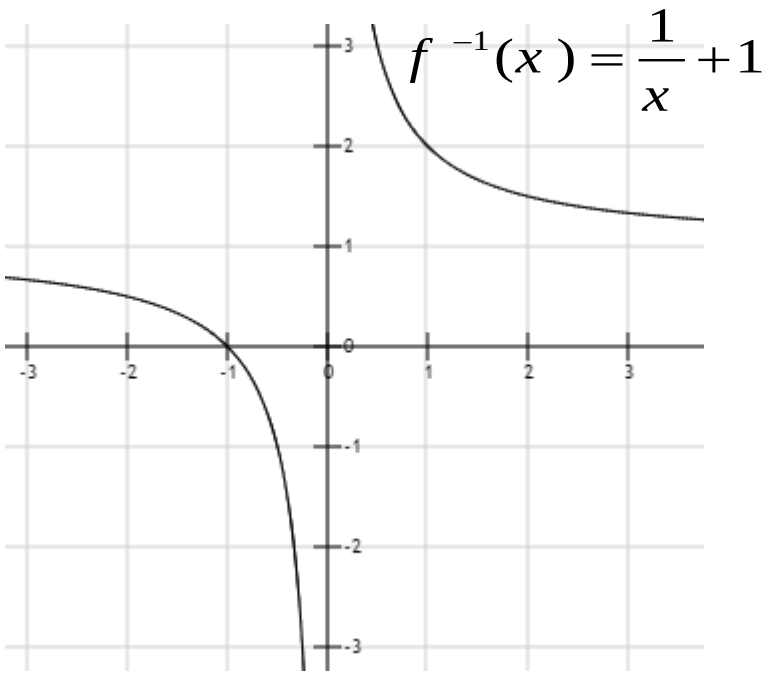
$$x = \frac{1}{y} + 1$$

$$f^{-1}(x) = \frac{1}{x} + 1$$

Finding Inverses: Example Continued

Example **Find the inverse function (if it exists) of $f(x) = \frac{1}{x-1}$, sketch it , and write its domain and range**





Exponential Function

Exponential Function

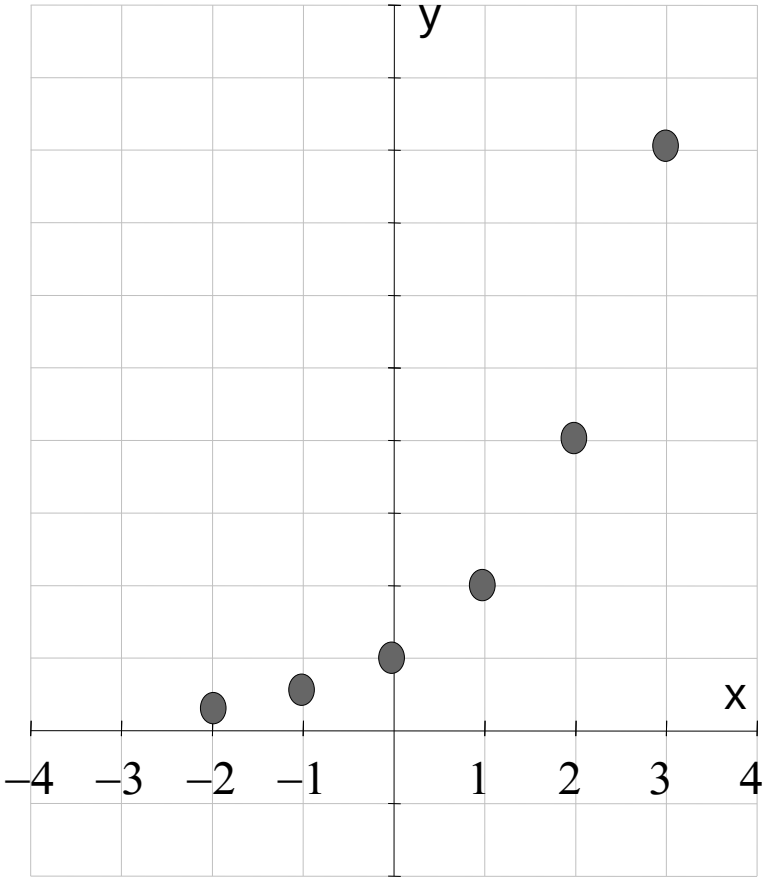
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$$y = (const)^x$$

$$y = (3)^x, \; y = (2)^x$$

$$y = a^x, \quad a > 0, \; a \neq 1$$

$$y = 2^x$$



Exponential Function

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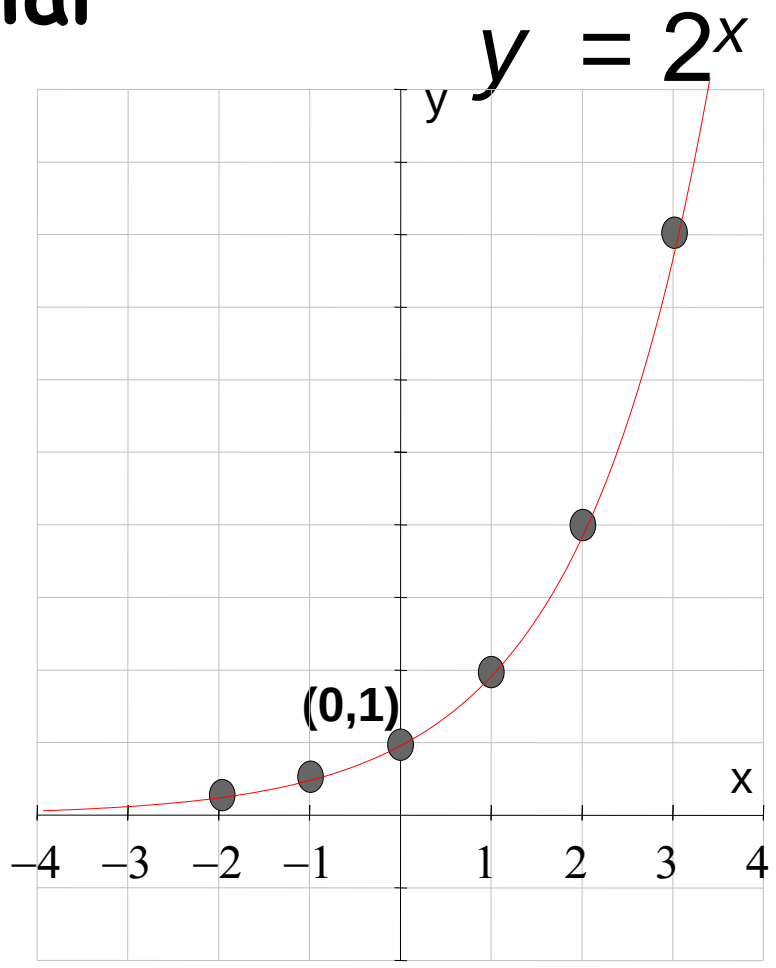
- The values of **Exponential Function** always positive

- *No intersection with* *x*-axis

$$D(2^x) = \Re$$

$$\mathrm{R}(2^x) = (0, \infty)$$

Neither even nor odd



Exponential Function

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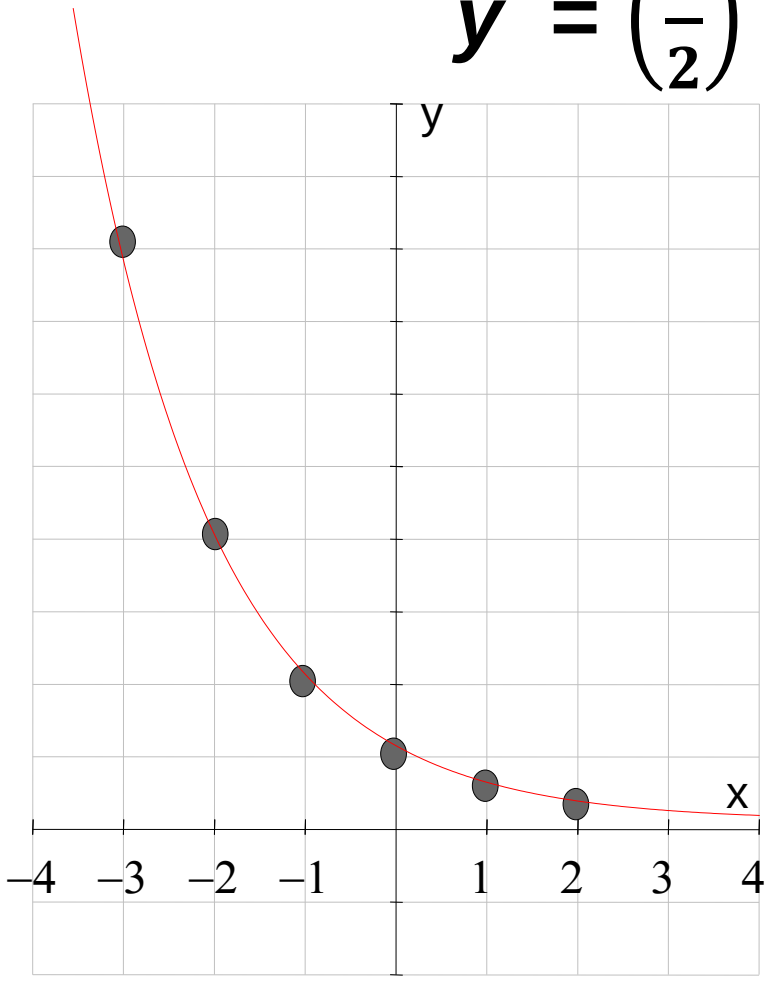
$$y = \left(\frac{1}{2}\right)^x$$

- The values of **Exponential Function** always positive

- *No intersection with x-axis*

$$D\left((0.5)^x\right) = \Re$$

$$\mathrm{R}\left((0.5)^x\right) = (0, \infty)$$



Exponential Function

$$y = a^x, \quad a > 0, \quad a \neq 1$$

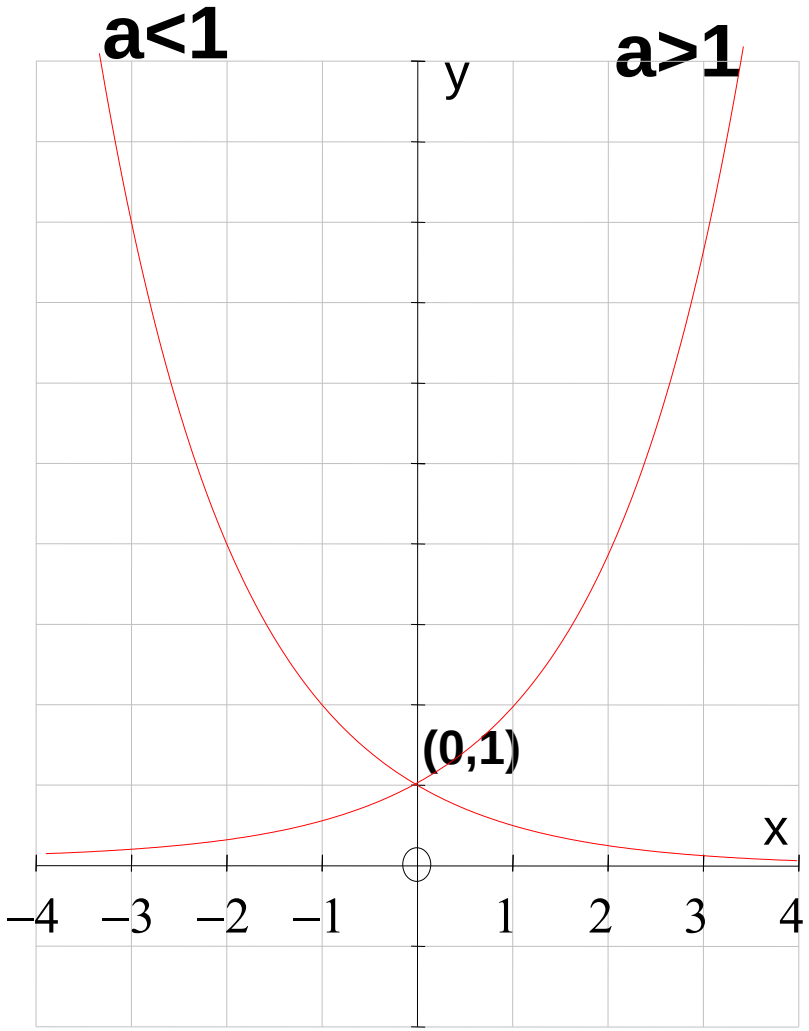
$$D(a^x) = \Re$$

$$R(a^x) = (0, \infty)$$

Rules: If $\mathbf{a, b > 0}$

$$a^x \cdot a^y = a^{x+y}$$

$$\frac{a^x}{a^y} = a^{x-y} \qquad \left(a^x\right)^y = a^{xy}$$



Natural Exponential Function

The natural exponential function is a special case of the exponential function

$$y = e^x \qquad e = 2.718281 > 1$$



- e is called **Euler's number.**
- e is a famous irrational **number**.

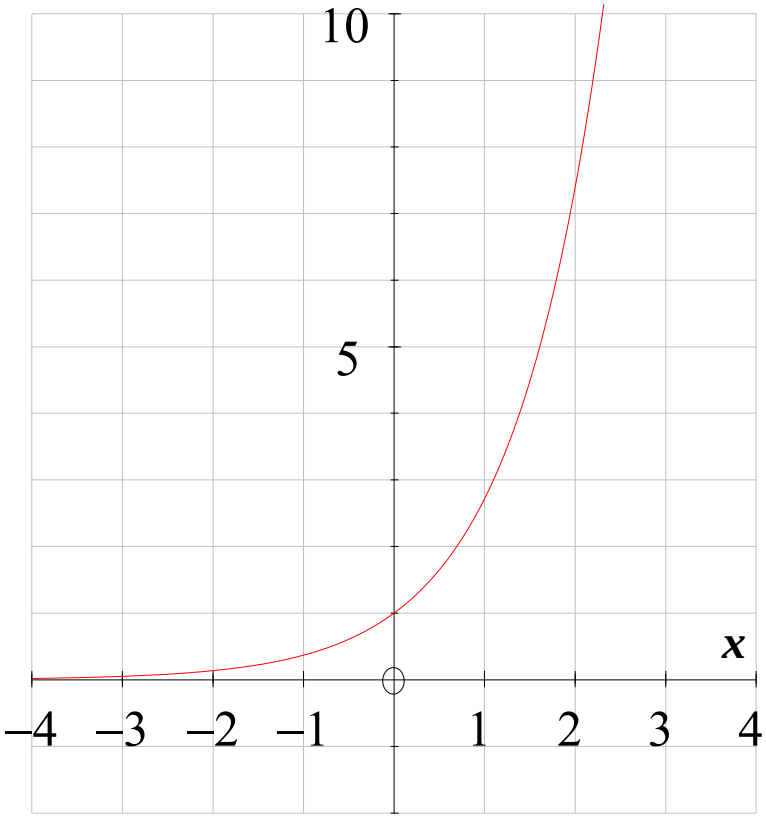
Natural Exponential Function

$y = e^x, \quad e = 2.7182818 > 1$

$D(e^x) = \Re$

$R(e^x) = (0, \infty)$

$y = e^x$



logarithmic Function

Logarithmic function

power

$$y = \log_a x. \quad \Leftrightarrow \quad a^y = x$$

base

Example: find value of each of the following logarithms

$$\log_2 8 = 3$$

$$\log_3 \left(\frac{1}{9}\right) = -2$$

$$\log_3 1 = 0$$

$$\log_a (0) = \text{Undefined}$$

$$\log_a (-1) = \text{Undefined}$$

Logarithmic function

$f\ (f^{-1}(x))\ =\ x$, $f^{-1}(f(x))=\ x.$

$\log_a(a^x)=x$ $a^{\log_a(x)}=x$

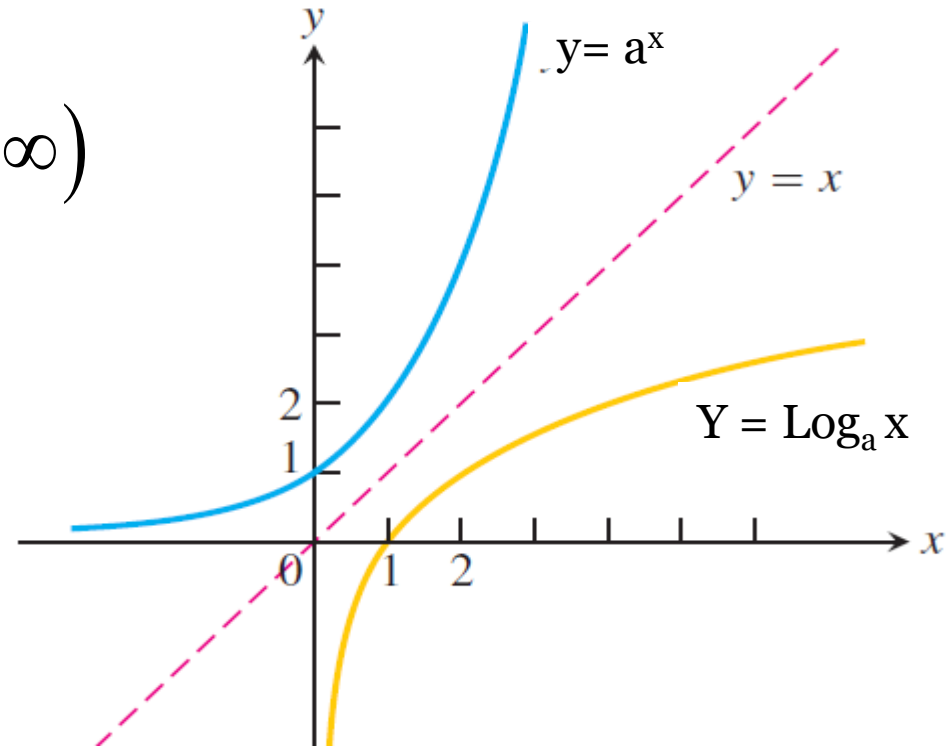
Example: solve

$\log_3(3^4)=\ \ 4$ $3^{\log_3(9)}=\ 9$

Graph: Logarithmic function

The domain is $(0, \infty)$

The range is $(-\infty, \infty)$



Natural Logarithmic function

Special case of Logarithmic function $y = \log_e x = \ln x.$

Inverse of natural exponential function

$$e^{\ln(x)} = x \qquad \ln(e^x) = x$$

Example: solve

$$e^x = 3$$

Taking $\ln$ for both sides

$$\ln(e^x) = \ln(3)$$

$$x = \ln(3)$$

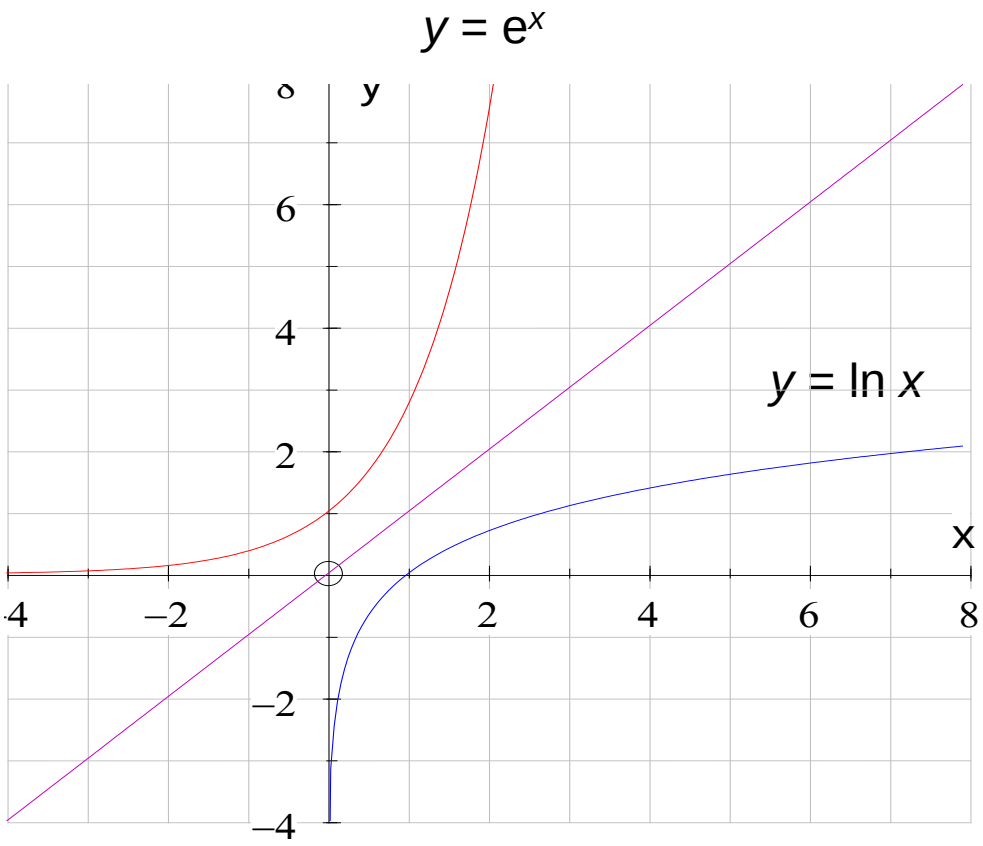
$$\ln(x) = 5$$

Take e for both sides

$$e^{\ln(x)} = e^5$$

$$x = e^5$$

Natural Logarithmic function graph



$$D(\ln x) = (0, \infty)$$

$$R(\ln x) = \Re$$

$$\ln(\infty) = \infty$$

$$\ln(0) = -\infty$$

Example: Find the domain

1) $f(x) = \ln(x-3)$

$x-3>0$

$x>3$

$D(f) = (3,\infty)$

2) $f(x) = \ln|x|$

$|x| > 0$

$D(f) = \Re - \{0\}$

3) $f(x) = \ln(\ln(x))$

$x>0$ | $(\ln(x)) > 0$

$x > e^0$

$x>1$

$D(f) = (1,\infty)$

Logarithmic Function Rules

For $x > 0$, $y>0$, a>0, $a \neq 1$, and any real number r,

Product Rule $\log_a xy = \log_a x + \log_a y$

Quotient Rule $\log_a \frac{x}{y} = \log_a x - \log_a y$

Power Rule $\log_a x^r = r \log_a x$

Same rules apply to the natural logarithmic function

$$\ln(xy) = \ln x + \ln y$$

$$\ln \frac{x}{y} = \ln x - \ln y$$

$$\ln x^r = r \ln x$$

Derivatives

Derivative of Power Functions

$$f(x) = x^n, \quad n \in R \quad \Rightarrow \quad f'(x) = nx^{n-1}$$

Example: Find $f'(x)$ if

$$f(x) = x^3 + x^{4/3} + \frac{1}{\sqrt{x}}$$

$$f'(x) = 3x^2 + \left(\frac{4}{3}x^{1/3}\right) + \left(\frac{-1}{2}x^{-3/2}\right)$$

Differentiation Rules

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If $f(x)$ and $g(x)$ are differentiable functions and c is a constant, then

$$i)\ \frac{d}{dx}(c) = 0$$

$$ii)\ \frac{d}{dx}\left(cf(x)\right) = c\frac{d}{dx}\left(f(x)\right) = cf'(x)$$

$$iii)\ \frac{d}{dx}\left(f(x) + g(x)\right) = f'(x) + g'(x)$$

Differentiation Rules

If $f(x)$ and $g(x)$ are differentiable functions and c is a constant, then

$$iv)\ \frac{d}{dx}\left(f(x)\times g(x)\right)=f'(x)g(x)+f(x)g'(x)$$

$$v)\ \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)=\frac{f'(x)g(x)-f(x)g'(x)}{g^2(x)},\quad g(x)\neq 0$$

Differentiation Rules

Examples:- Find the derivative of the following functions

$$i)\, y = \left(x^2 - x + 5\right)\left(x^3 + 4x - 1\right)$$

$$y' = (2x-1)\left(x^3 + 4x - 1\right) + \left(x^2 - x + 5\right)\left(3x^2 + 4\right)$$

$$ii)\, y = \frac{x^2 + 3}{4x - 1}$$

$$y' = \frac{2x(4x-1) - 4\left(x^2 + 3\right)}{(4x-1)^2}$$

Derivatives of Trigonometric Functions

Function	Derivative	Function	Derivative
$\sin(x)$	$\cos(x)$	$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$	$\cot(x)$	$-\csc^2(x)$
$\sec(x)$	$\sec(x)\tan(x)$	$\csc(x)$	$-\csc(x)\cot(x)$

Derivatives of Trigonometric Functions

Example: Find the derivative of the following functions

$$i)\ f(x) = x^2 \sin x$$

$$f'(x) = 2x \sin x + x^2 \cos x$$

$$ii)\ g(x) = \frac{\cos x}{x^3}$$

$$g'(x) = \frac{x^3(-\sin x) - \cos x\ (3x^2)}{(x^3)^2}$$

Derivatives of Exponential and Logarithmic Functions

Function	Derivative	Function	Derivative
e^x	e^x	$ln(x)$	$\frac{1}{x}$
a^x	$a^x \ln(a)$	$\log_a(x)$	$\frac{1}{x\ln(a)}$

Derivatives of Exponential and Logarithmic Functions

Example **Find the derivative of the following functions**

i) $y = 7^x$ $\qquad y' = 7^x \ln 7$

ii) $y = x\, e^x$ $\qquad y' = (1)e^x + xe^x$

iii) $y = 7^x + x^7$ $\qquad y' = 7^x \ln 7 + 7x^6$

The Chain Rule

If $y = f(u)$ and $u = g(x)$ then $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

$$\left[f(g(x)) \right]' = f'(g(x))g'(x)$$

let $y = \sin u$, $u = x^2$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = (\cos u)(2x) = \left(\cos x^2\right)(2x)$$

$$\therefore \quad \frac{d}{dx}\sin(x^2) = \left(\cos x^2\right)(2x)$$

Outside/Inside method

The Chain Rule

Example **Find the derivative of the following functions**

$$y = \cos\left(x^2 - 4\right)$$

$$y' = \left(-\sin\left(x^2 - 4\right)\right)\ (2x)$$

$$y = \sec\left(e^x\right)$$

$$y' = \sec\left(e^x\right)\tan\left(e^x\right)\left(e^x\right)$$

The Chain Rule

Example **Find the derivative of the following functions**

$$y = e^{\sin(2x+1)}$$

$$y' = e^{\sin(2x+1)} \cos(2x+1)\ (2)$$

$$y = \sin\left(\sin(\sin x)\right)$$

$$y' = \cos\left(\sin(\sin x)\right)\ \left(\cos(\sin x)\right)\quad (\cos x)$$

The Chain Rule

Note: $y = 10^x$ $\quad y' = 10^x \ln 10$

Example **Find the derivative of the following function**

$$y = 10^{\sin 3x}$$

$$y' = 10^{\sin 3x} (\ln 10) \quad (\cos 3x) \quad (3)$$

The Chain Rule

Example **Find the derivative of the following functions**

$$y = x\ln(\sin x)$$

$$y' = (1)\ln(\sin x) + x\frac{\cos x}{\sin x}$$

$$y = \left(1 + 2x^3 - \sec x\right)^4$$

$$y' = 4\left(1 + 2x^3 - \sec x\right)^3 \left(6x^2 - \sec x \tan x\right)$$

Implicit Differentiation

- **Explicit functions**

$$y = 3x - 2$$

$$y = x^2 + 5$$

- **Implicit functions**

$$y^2 + 2yx + 4x^2 = 0$$

$$y^5 - x^2 \sin y + 2 = 0$$

Implicit Differentiation

$$\frac{d}{dx}x = 1 \qquad \frac{d}{dx}y = y'$$

$$\frac{d}{dx}\sin x = \cos x \qquad \frac{d}{dx}\sin y = (\cos y)y'$$

$$\frac{d}{dx}x^2 = 2x \qquad \frac{d}{dx}y^2 = 2yy'$$

$$\frac{d}{dx}e^x = e^x \qquad \frac{d}{dx}e^y = e^y y'$$

$$\frac{d}{dx}\ln x = \frac{1}{x} \qquad \frac{d}{dx}\ln y = \frac{1}{y}y'$$

Implicit Differentiation: Example

Example Find y' if $2y = x^2 + \sin y$

Differentiate with respect to x

$$2y' = 2x + \cos y\, y'$$

$$2y' - \cos y\ y' = 2x$$

$$y'(2 - \cos y) = 2x$$

$$y' = \frac{2x}{2 - \cos y}$$

Logarithmic Differentiation: Example

Example **Find** y' **if** $y = \ln\left(x^3 \sin^2 x \sqrt{1+x}\right)$

$$y = \ln(x^3) + \ln(\sin^2 x) + \ln\left(\sqrt{1+x}\right) \quad \text{Properties of ln}$$

$$y = 3\ln(x) + 2\ln(\sin x) + 0.5\ln(1+x) \quad \text{Properties of ln}$$

$$y' = 3\frac{1}{x} + 2\frac{\cos x}{(\sin x)} + 0.5\frac{1}{1+x}$$

If $y = x^3 \sin^2 x \sqrt{1+x}$

take ln to both sides to make it easier to solve

Logarithmic Differentiation: Example

Example **Find** y' **if** $y = \dfrac{(x^2-3x+4)^5\sqrt{\cos 6x}}{e^{7x+1}*\ln(8x-9)}$

$$\ln y = \ln(x^2-3x+4)^5 + \ln\sqrt{\cos 6x} - \ln(e^{7x+1}) - \ln(\ln(8x-9))$$

$$\ln y = 5\ln(x^2-3x+4) + \frac{1}{2}\ln(\cos 6x) - (7x+1) - \ln(\ln(8x-9))$$

$$\frac{y'}{y} = 5 * \frac{2x-3}{x^2-3x+4} + \frac{1}{2} * \frac{-6\sin 6x}{\cos 6x} - 7 - \frac{1}{\ln(8x-9)} * \frac{8}{8x-9}$$

$$y' = y\left(\frac{5(2x-3)}{x^2-3x+4} - 3\tan 6x - 7 - \frac{8}{(8x-9)\ln(8x-9)}\right)$$

Logarithmic Differentiation: Example

Example **Find the derivative of the following functions**

$$(\mathrm{i}) y = x^x$$

$$\ln y = \ln x^x = x \ln x$$

$$\frac{y'}{y} = (1)(\ln x) + x\left(\frac{1}{x}\right) = 1 + \ln x$$

$$y' = y(1 + \ln x) = x^x(1 + \ln x)$$

$$(\mathrm{ii}) y = x^{\sin(x)}$$

Try to solve

See you next week



